

Seminars 9 & 10: IIR Filters

Topics: recursive difference equations, initial-rest conditions, impulse and step responses, rational system functions, direct-form structures, regions of convergence, poles and zeros, stability, frequency response, inverse z-transforms, transient and steady-state behaviour, and second-order IIR filters.

Key idea. IIR filters use feedback, so their responses can continue after the input has ended. The poles of the rational system function govern transient behaviour and stability, while evaluation on the unit circle gives the frequency response.

1. Recursive difference equations and initial rest

Consider the first-order recursive system $y[n] = 0.8y[n-1] + x[n]$ with input $x[n] = u[n]$ and initial-rest conditions.

- (a) State explicitly what initial-rest conditions mean for $x[n]$ and $y[n]$.
- (b) Compute $y[0], y[1], y[2], y[3]$ directly from the recursion.
- (c) Write $y[n]$ for general $n \geq 0$ as a finite geometric sum.
- (d) Use the geometric-series formula to obtain a closed-form expression for $y[n]$.
- (e) Determine the final value approached by the step response and explain why it is finite.

2. Impulse response of a first-order IIR filter

Consider $y[n] = 0.6y[n-1] + 2x[n]$

- (a) Apply $x[n] = \delta[n]$ and calculate the first five samples of the impulse response $h[n]$.
- (b) Derive a closed-form expression for $h[n]$ valid for all n .
- (c) Explain why this is an infinite impulse response even though the input impulse has duration one sample.
- (d) Is the impulse response absolutely summable? What does your answer imply about stability?
- (e) Repeat qualitatively for a recursive coefficient of 1.05 and compare the two responses.

3. System function from a recursive difference equation

Consider the first-order IIR filter $y[n] = 0.7y[n-1] + x[n] - 0.4x[n-1]$

- (a) Take the z-transform under initial-rest conditions and derive $H(z) = Y(z)/X(z)$.
- (b) Identify the numerator coefficients b_0, b_1 and denominator coefficients.
- (c) Determine the pole and zero locations.
- (d) Sketch the pole-zero diagram and the unit circle.
- (f) Explain which part of the structure is feed-forward and which part is feedback.

4. Region of convergence of an exponential sequence

Consider the causal sequence $x[n] = (0.75)^n u[n]$

- (a) Starting from the z-transform definition, show that the transform is a geometric series.
- (b) Find $X(z)$ in closed form.
- (c) Determine the region of convergence (ROC).
- (d) Locate the pole in the z-plane and indicate the ROC on the same sketch.
- (e) Does the ROC include the unit circle? Explain the significance of this for the existence of the frequency response.

5. Poles, zeros, and stability

For each of the following causal system functions, find the poles and zeros and determine whether the system is stable: $H_1(z) = 1/(1 - 0.8z^{-1})$, $H_2(z) = (1 + z^{-1})/(1 - 1.1z^{-1})$, $H_3(z) = (1 - 0.5z^{-1})/(1 + 0.6z^{-1})$

- (a) Plot all pole-zero locations in the z -plane.
- (b) For each system, state the causal ROC.
- (c) Apply the pole-location stability criterion.
- (d) Describe the expected long-time behaviour of each impulse response.
- (e) Explain why a pole close to the unit circle produces a long transient.

6. Frequency response of a first-order IIR filter

Consider $H(z) = (1 + z^{-1})/(1 - 0.8z^{-1})$

- (a) Determine the pole and zero locations and verify stability.
- (b) Obtain $H(e^{j\hat{\omega}})$ by substituting $z = e^{j\hat{\omega}}$.
- (c) Evaluate the magnitude at $\hat{\omega} = 0, \pi/2, \pi$.
- (d) Evaluate or sketch the phase at the same frequencies.
- (e) Explain why the response is large near DC and zero at the Nyquist frequency.
- (f) For input $x[n] = \cos(\pi n/2)$ that exists for all n , write the steady-state output.

7. From a rational system function to the impulse response

Consider the causal system $H(z) = (1 - 0.2z^{-1})/(1 - 0.6z^{-1})$

- (a) Rewrite $H(z)$ as a direct term plus a first-order rational term.
- (b) Use a known z -transform pair to obtain $h[n]$.
- (c) Verify your result by computing $h[0], h[1], h[2]$ from the difference equation.
- (d) Explain how the numerator changes the first part of the impulse response while the pole controls its exponential tail.

8. Inverse z -transform by partial fractions

Suppose $Y(z) = 1/((1 - 0.5z^{-1})(1 - 0.8z^{-1}))$

- (a) Find the two poles.
- (b) Perform a partial-fraction expansion of $Y(z)$.
- (c) Assuming a causal sequence, compute the inverse z -transform $y[n]$.
- (d) Calculate the first four samples from your closed-form result.
- (e) Which pole dominates the response for large n and why?

9. Transient and steady-state response to a switched-on sinusoid

Consider the stable first-order filter $y[n] = 0.85y[n - 1] + x[n]$ with $x[n] = \cos(0.3\pi n)u[n]$.

- (a) Explain why the response contains both a pole-dependent transient and a sinusoidal steady-state component.
- (b) Determine $H(e^{j0.3\pi})$ and hence the steady-state sinusoidal response.
- (c) State the general form of the transient component.
- (d) Does the transient decay or grow? Explain from the pole location.
- (e) Approximately how many samples are needed for the transient amplitude to fall below 10% of its initial value?

10. Second-order poles and the difference equation

A second-order system has poles $p_{1,2} = 0.9e^{\pm j\pi/4}$

- (a) Form the denominator polynomial in z^{-1} with real coefficients.
- (b) Write a second-order recursive difference equation with numerator $B(z) = 1$.
- (c) Plot the two poles and the unit circle.
- (d) Is the causal system stable?
- (e) State the oscillation frequency and explain the role of the pole radius in the transient envelope.

11. Impulse response of a second-order resonator

Consider $H(z) = 1/(1 - 1.6\cos(0.2\pi)z^{-1} + 0.64z^{-2})$

- (a) Factor the denominator and identify the two poles.
- (b) Determine their radius and angle.
- (c) Predict the qualitative shape of the causal impulse response before calculating it.
- (d) Use partial fractions or the standard conjugate-pole form to express $h[n]$ as a damped sinusoid.
- (e) What would change if the pole radius were moved from 0.8 to 1.02 without changing the angle?

13. Connecting the n , z , and frequency domains

For the causal system $H(z) = (1 - z^{-1})/(1 - 0.9z^{-1})$

- (a) Write the difference equation.
- (b) Determine the impulse response.
- (c) Find the pole and zero and discuss stability.
- (d) Evaluate the frequency response at $\hat{\omega} = 0$ and $\hat{\omega} = \pi$.
- (e) Explain how the zero at DC and the pole near $z = 1$ together shape the frequency response.
- (f) Summarize how the same filter is represented in the time domain, z domain, and frequency domain.

14. Designing an IIR oscillator from desired poles

You want the impulse response of a second-order recursive system to contain a damped oscillation at normalized frequency $\hat{\omega}_0 = 0.25\pi$ with an envelope that decreases by a factor of 0.95 per sample.

- (a) Choose the required complex-conjugate pole pair.
- (b) Derive the real denominator polynomial.
- (c) Write a second-order difference equation with $B(z) = 1$.
- (d) Is the system stable?
- (e) How would the sound or waveform change if the radius were increased to 0.995 while the pole angle stayed fixed?
- (f) How would it change if the angle were changed to 0.4π while the radius stayed 0.95?

15. Computational investigation with Python

Use NumPy, Matplotlib, and scipy.signal to investigate IIR filters.

- (a) For first-order systems $H(z) = 1/(1 - az^{-1})$ with $a = 0.5, 0.9, 0.99, 1.05$, plot the impulse responses and pole locations.
- (b) For each case, classify stability and relate the pole radius to the decay or growth rate.

- (c) Use `scipy.signal.freqz` to plot the magnitude and phase of a stable first-order system and verify the response by filtering a sinusoid.
- (d) For a second-order system with poles $0.95e^{\pm j0.3\pi}$, plot the impulse response and confirm that the oscillation frequency agrees with the pole angle.
- (e) Use `scipy.signal.lfilter` to compare direct recursion with the output predicted by convolution with a long truncated impulse response.
- (f) Summarize how recursion, poles, stability, transients, and frequency response are related.